

Investment Opportunities, Liquidity Premium and Conglomerate Mergers

Chun Chang and Xiaoyun Yu¹

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¹Chang is at the Carlson School of Management, University of Minnesota and Yu is at the Kelley School of Business, Indiana University. This paper is a substantial revision of our earlier paper entitled “Investment Opportunities and the Informational Benefits and Costs in Conglomerate Mergers”. We have benefited enormously from the comments and suggestions made by an anonymous referee. We also thank Larry Benveniste, John Boyd, Mike Fishman, Scott Gibson, Vojislav Maksimovic, Paul Seguin, and Andy Winton for helpful comments and suggestions. We are especially grateful to Jack Kareken for his help in re-writing much of the paper to improve its readability. All errors are our own responsibility. Chang acknowledges financial support from Minnesota Banking Industry Professorship Endowment, and Yu thanks the Center for Entrepreneurial Studies at the University of Minnesota for research support.

Abstract

This paper analyzes the implications of diversification strategies and conglomerate mergers for the cost of capital in a model where the cost of capital depends on liquidity. We show that in a finitely liquid market with asymmetrically informed investors, both the benefits and the costs of diversification can be related to two characteristics of a firm's investment opportunities. The benefits come from a reduced liquidity discount in the stock price of the merged firm when its shareholders anticipate less informed trading. The costs are the result of less efficient investment by its divisions due to a less informative stock price. The benefits and the costs vary with the return and risk of the investment opportunities of the firm's divisions. Our results can explain the life cycle of diversification strategies that many firms experience. They also provide new implications for evaluating merger and spin-off candidates, and for understanding the trade-off between internal and external capital markets for capital allocations.

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1 Introduction

The extent to which a firm should diversify is a central issue in business organization. Although “focus” has become a fashionable word in management circles and on average diversified U.S. firms trade at a discount relative to their focused counterparts,¹ a large fraction of diversified firms (around 40% according to Denis, Denis and Sarin 1997) still trade at premiums. Researchers have also shown that the conglomerate mergers created value historically and that there is no or less diversification discount for firms in other countries.² It is therefore important to identify the economic determinants of mergers and spin-offs that are capable of explaining why the conglomerate discount varies over time and across industries and countries.

In this paper, we explore the relationship between a firm’s diversification strategy and the liquidity premium in its stock return. We believe this issue is potentially very important for the following reasons. First, the liquidity premiums due to adverse selection problems in security trading are shown to have a significant impact on stock returns. Brennan and Subrahmanyam (1996) have shown that the return difference between the most liquid shares and the least liquid ones on the NYSE is 6.62 percent per year.³ If the liquidity premium can be reduced by only a few percentage points, huge gains in valuation can result.⁴ Second, by relying on changes in the two key characteristics of investment opportunities, the theory developed here is capable of explaining the life cycle of diversification strategies for many firms: from focus to diversification and then back to focus again. Third, the merger benefits identified here are the same ones as those that presumably give rise to the popularity of stock index futures and asset securitization (Gorton and Pennacchi 1990, Subrahmanyam 1991, Boot and Thakor 1993). To the extent that these financial innovations created economic benefits, so should conglomerate mergers.

¹See Lang and Stulz (1994), Berger and Ofek (1995), and Servaes (1996).

²See, for example, Schipper and Thompson (1983), Matsusaka (1993), Hubbard and Palia (1998), Lins and Servaes (1998), and Fauver, Houston and Naranjo (1999).

³Note that the most liquid stocks tend to be those of larger, more diversified firms whereas the least liquid ones tend to be those of smaller, more focused firms. The return difference would be even larger if very small and highly focused firms were listed on the NYSE.

⁴By way of illustration, we provide an example using reasonable numbers. Assume that a firm’s equity rate of return or discount rate is 12% and its (constant) dividend growth rate is 6%. With a three percentage point reduction in the discount rate, the share price doubles; with a four percentage point reduction, the price triples.

Mergers may be the only way to realize the benefits when, and in countries where, the new financial instruments were less developed.

Before describing our model, it is useful to distinguish between two types of information flows between firms and the capital market. First, information flows from firms to the capital market through managerial actions when managers possess information that the market does not have. This perspective has produced many important insights in finance in the past two decades. On the other hand, it has long been recognized that information also flows from markets to firms (Hayek 1945). In particular, security prices can aggregate decentralized and dispersed information in the economy and provide important signals for firms' investment decisions. The information that flows from the market to firms can be referred to as "allocative information".⁵ Empirical evidence in the IPO literature is consistent with the view that the IPO process generates considerable amount of information about IPO firms' growth potentials and values that is not in the security issuers' information set. In recent years, an emerging literature in finance has recognized the importance of allocative information. For example, its efficient production has been viewed as a key factor when comparing financial systems in different countries (Allen 1993, Boot and Thakor 1997, Dow and Gorton 1997, Rajan and Zingales 1998, and Subrahmanyam and Titman 1999).

Our model is built on this second type of information flow. To understand the connection between the allocative information and conglomerate mergers, we construct a model of how mergers affect the way in which information about business units' investment opportunities is conveyed by stock prices in the secondary market and how the information is then utilized in making investment decisions. In other words, what we have is a market microstructure model in which the firm's optimal investment decision is part of the equilibrium. Such a model, while very difficult to analyze, allows us to identify both the benefits and the costs of generating the allocative information. These benefits and costs are closely related to the characteristics of the firm's investment opportunities, which we believe to be one of the major determinants of mergers.

⁵The usefulness of the allocative information for firms' investment decisions can be motivated either by the fact that managers does not have it or by agency considerations. See next section for details.

Specifically, we investigate how a conglomerate merger of two unrelated business units affects speculators' incentives to collect information about the business units' investment opportunities. We characterize the conditions under which the speculators will collect information about the units with and without a merger. We show that the speculators are less likely to collect information on both units after a merger. This becomes an economic cost of a conglomerate merger if the information is valuable for making investment decisions. On the other hand, less informed trading reduces the loss of average shareholders when they have to trade for liquidity reasons. Consequently, average shareholders are willing to pay a higher price for the firm initially. The liquidity premium required by them is therefore smaller. This is the benefit of a conglomerate merger.⁶

The discussion above makes it clear that the cost and the benefit of a merger depend on whether the allocative information is valuable. If there is little uncertainty about the NPV of a business unit's project (the variance is small compared to the mean), the social value of the allocative information to the firm is small. The firm should discourage informed traders to produce this information because they gain at the expense of the uninformed investors. This can be achieved through a conglomerate merger. The model allows us to explicitly compute firms' values under mergers and spin-offs as functions of the means and variances of the NPVs of the two business units' investment opportunities and the extent of investors' liquidity shocks.

By linking the benefit and cost of a merger to the risk and return of the firm's investment opportunities, our model can explain the life cycle of diversification strategies that many firms experience. When a young firm such as a new technology firm starts out, the uncertainty of its projects' NPV is usually very high compared to its mean. As the firm becomes more established, the uncertainty of its NPV will decline compared to its mean. This corresponds to the stage where the demand for its product is better established and there is still little competition. Firms such as Cisco, Intel and Microsoft are somewhere at this stage. As the firm matures and more competitors enter into its business, the mean of its NPV declines relative to the uncertainty. That is, many firms' investment opportunities change

⁶In recent empirical work, Huson and MacKinnon (2001) present evidence that is consistent with increased informed trading following corporate spin-offs.

according to the following pattern: from high return high risk, to high return low risk, and then to low return high risk. Our model then predicts that they will start as more focused business, then diversify into other businesses, and finally return to become more focused businesses again. This seems to be consistent with the life cycle of diversification strategies that many firms follow.⁷

Habib et. al. (1997) first formalized the idea that spin-offs can improve the efficiency of investment decisions when some traders have superior allocative information that managers do not have. They use a rational expectation equilibrium setup in which the price formation process is not explicitly modeled. The informed traders take stock prices as given and behave competitively. There is no cost for stock prices to convey allocative information. Consequently, spin-offs are always preferred to mergers in their model. Chang (1998) showed that spin-offs can increase one type of information flow at the expenses of another type of information flow. The paper does not model the price formation process either. The trade-off is driven by agency considerations and is therefore quite different.

A number of papers have studied the relationship between market liquidity and the efficiency of decision making at the firm's level. Holmström and Tirole (1993) show that low market liquidity induced by ownership concentration reduces market monitoring of managerial performance, which in turn lowers the level of managerial effort. Kahn and Winton (1998) and Maug (1998) investigate how market liquidity affects large shareholders' incentive to intervene in poor performing firms. These papers focus on the effects of ownership concentration on stock price liquidity. In contrast, we study the effects of mergers and spin-offs. The issue here is more difficult because we have to deal with the revelation of two types of information and examine the efficiency of investment decisions of both firms (or business units). Subrahmanyam and Titman (1999) studied how information revealed in the stock prices can improve firms' investment decisions. They are mainly interested in the comparison between intermediated financing by banks and direct financing by markets.

Although it was proposed long time ago that internal capital markets can have informational advantage (Alchian 1969 and Williamson 1970), their benefits and costs have mainly been analyzed from

⁷Although a rigorous empirical test is well beyond the scope of this paper, one way to empirically measure the two characteristics is to use the mean and the dispersion of financial analysts' growth rate forecasts.

the perspective of the incomplete contracts (Gertner, Scharfstein and Stein 1994 and Stein 1997). The informational benefits and the costs of conglomerate mergers in our model can also be viewed as those of internal capital markets. Our results are consistent with the empirical evidence that investment decisions made in internal capital markets are less efficient (Lamont 1997 and Shin and Stulz 1998). Moreover, our detailed analysis in section 4 provides specific predictions on what types of divisions are likely to make inefficient investment decisions.

Relationships between business units are much more complicated than those captured in our model. Many other factors are important. Some of these factors have been discussed in the literature on the boundaries of the firm.⁸ Our focus on the informational benefits and costs does not mean that they are the only important determinants of conglomerate mergers.

The paper is organized as follows. The next section lays out the structure of the model. In section 3, we derive and analyze the equilibria when the two businesses units are separately traded (the spin-off case). In section 4, we do the same when the two units are integrated and issue a single stock (the merger case). Section 5 derives explicit conditions under which a merger is optimal and the associated economic implications. Section 6 discusses the implications of variations of the model and concludes the paper. All proofs are in the appendix.

2 The Model

Models that allow explicit analyses of the price formation process tend to be complex. In this paper, a new dimension (the investment decision) is added to the standard trading models. In order to be tractable, we have to simplify everything that is not essential to the issues in which we are interested. It will become clear that the main result that merger gains depend on the two characteristics of the investment opportunities does not rely on the many simplistic features of the model.

⁸See Hart (1989), Holmström and Tirole (1989), and Milgrom and Roberts (1992) for more references.

2.1 The Timing of Events

We consider two unrelated business units M and N . There are three dates. At date 0, they can be integrated into one firm whose shares are sold to investors. Alternatively, the two units can be organized as two independent firms and each firm's shares are sold to investors. We will refer to the first arrangement as merger (or integration) and the second arrangement as spin-off (or disintegration).⁹ As in Kyle (1985), we assume that there is one trader who can collect costly information about the two business units. Whether the trader will collect information on one or both units will depend on the organizational arrangement.¹⁰

At date 1, the investors who bought shares at date 0 experience liquidity shocks and a random fraction of them sell their shares. The informed trader can also submit trading orders. As in Kyle (1985), the orders from liquidity traders and the informed trader will be sent to market makers who set a competitive price based on the total order submitted for a stock. When the two units are disintegrated, the informed trader can trade in both firms' stock, and there will be two stock prices.

After trading at date 1, each business unit makes its investment (or liquidation) decisions based on the information conveyed by the stock price(s). At date 2, the payoffs of the business units are realized and paid to shareholders.

The details of the model and the assumptions are discussed below.

2.2 Investment Technology

The NPV of a business unit's investment opportunity is uncertain. We would like to vary both the mean and dispersion of the investment opportunity. We assume that the NPV of unit M 's investment project is $m + \varepsilon$, where ε is either e or $-e$ with equal probability 0.5. The NPV of unit N 's investment

⁹Other methods of divesting businesses such as equity carve-outs or issuing tracking stocks also achieve the goal of having a business unit separately traded. We believe, however, the divestiture method in our paper is best interpreted as spin-offs for the following reasons. In order to capture the essence of tracking stocks, one needs to distinguish between earnings and assets. We are unable to accomplish this in our one-period model. In equity carve-outs, the divesting firm typically retains a majority stake. We do not consider the role of large shareholders in our model.

¹⁰We have also considered the case where there are two traders specializing in information collection: each trader can become informed about only one business unit. These results are available upon request.

project is $n + \delta$, where δ is either d or $-d$ with equal probability 0.5. Without loss of generality, we assume $\inf \{m, n\} \geq 0$.

If we were to replace the term “invest” with “continue” and “not invest” with “discontinue”, the investment decision could be interpreted as a continuation decision.

2.3 The Informed Trader $\mathcal{I}$

As discussed above, we assume that there is only one trader, $\mathcal{I}$, who can choose to become informed. To capture the decentralized nature of information, the identity of this trader is not known at date 0 when the organization structure is determined. That is, at date 0, neither of the two business units knows who has the access to the most accurate information needed for making the investment decisions. This will prevent them from simply hiring the informed trader to run the firm.¹¹

By incurring a cost c , the trader $\mathcal{I}$ can learn the true realization of the NPV of a business unit’s project. If the trader $\mathcal{I}$ would like to know both units’ NPVs, he have to spend $2c$.¹² The trader is risk neutral and can submit buy or sell orders.

2.4 The Investors/Liquidity Traders

To capture the effects of liquidity on date 0 valuation, we model the liquidity traders as the investors who buy shares at date 0 and who may experience liquidity shocks at date 1, as in Diamond and Dybvig (1983) and Gorton and Pennacchi (1993). Therefore, we only consider negative liquidity shocks which prompt the investors to sell their shares.¹³ In particular, we assume that a random fraction β of the date 0 investors sell their shares at date 1. It should be noted that β is the aggregate (undiversifiable) liquidity shock for the economy. Diversifiable liquidity shocks at the individual level will be averaged out when a firm has enough shareholders. Consequently, the β that a firm (or each division after a

¹¹This assumption, however, is not crucial if whoever runs the firm has a tendency to “build an empire” as discussed below.

¹²For simplicity, even the values of the two types of information are different, the information acquisition costs are assumed to be the same. This assumption does not change our results qualitatively.

¹³Negative liquidity shocks are introduced as in Gorton and Pennacchi (1993) and Maug (1998) because from these works, we know it is easier to solve for equilibria in this setting. We see no reason why the main intuitions obtained would be affected if the positive liquidity shocks were introduced.

spin-off) faces is the same.

To make the model solvable, the fraction β of selling shareholders is assumed to be discrete as in Gorton and Pennacchi (1993) and Maug (1998). In particular the realization of β is either 0 or b with equal probability 0.5, where we assume $b \leq 1$. The investors are risk neutral and maximize discounted expected payoffs. Without loss of generality, the discount rate is assumed to be 0.

2.5 The Market Makers

The market makers are risk neutral agents who will set a competitive price for a stock after observing the total order submitted. They cannot separately observe the order sent by the informed trader and the orders by the liquidity traders.

2.6 The Business Unit Managers

We assume that the two business units, M and N , will use the information conveyed by their stock prices in making investment decisions. There are two justifications for why their managers cannot make the efficient investment decisions. First, the managers (at least the top corporate managers) may not have all the information about the future conditions in the business unit's input and output markets. Second, the managers may be well informed. However, due to the agency problem pointed out by Jensen (1986), they will prefer investment regardless of its NPV.¹⁴ In this case, the board of directors or the top management will use the information conveyed by stock prices to make or approve investment decisions.

2.7 Trading Mechanisms

The trading mechanisms in the market microstructure literature are of two types. Included in one type are variations on the mechanism of Glosten and Milgrom (1985). Included in the other are variations on the mechanism of Kyle (1985). In a Glosten-Milgrom trading mechanism, market makers quote

¹⁴For this to be the case, the managerial compensation schemes must be ineffective. One way it will be if the managers covered are wealth constrained. In order to induce efficient investment, the managers have to be given large stakes in their firms. This may, however, be too costly to investors.

prices and traders trade at the quoted prices. Although this trading mechanism is quite intuitive, it is difficult to analyze informed traders' dynamic trading strategies in this setting. In order to make a price at which the informed trader trades dependent on his trading volume, we used a Kyle type trading mechanism. As in Gorton and Pennacchi (1993) and Maug (1998), the information trader is allowed to submit both sell and buy orders. These orders are combined with liquidity traders' sell orders and the net order flow is then sent to the market makers. The market makers observe only the net order flow.¹⁵

2.8 The Trading-Investment Equilibrium

The Bayesian-Nash Equilibrium concept is used in this paper. It is essentially the same as that used in many of the market microstructure models but with one additional dimension: each business unit's investment decision is made based on the information conveyed in its stock price; and market makers' decisions rationally anticipate each business unit's optimal investment decision.

Specifically, the equilibrium is defined as:

Definition 1 *An equilibrium consists of trader $\mathcal{I}$'s decision rule for acquiring information, his trading strategy given the information he has acquired, the market makers' belief (posterior distribution of the true state given the order flows), and the firm's beliefs given market makers' prices such that:*

1. *the firm maximizes its value based on its beliefs;*
2. *given the firm's decision rule in 1, the market makers quote prices so that, based on their beliefs, they break even;*
3. *given 1 and 2, trader $\mathcal{I}$'s trading strategy maximizes his expected trading profit;*
4. *given 1, 2, and 3, the beliefs are consistent (satisfy Bays rule wherever applicable);*

¹⁵Different trading mechanisms may affect the informed trader's profit and the way in which information is revealed. To demonstrate that our general results are qualitatively unchanged under alternative trading mechanisms, we analyzed another trading mechanism in which the informed trader can only submit sell orders as the liquidity traders. The total of sell orders is then sent to the market makers. The two mechanisms produce essentially the same results. We will only present the results associated with the first trading mechanism.

5. *given 1, 2, 3, and 4, trader $\mathcal{I}$ acquires one or both types of information, his expected trading gain net of acquisition costs is maximized at date 0.*

The sequence of the events is given in Figure 1.

3 Liquidity, Investment Efficiency and Value under Spin-off

In this section, we consider the case where the two units, M and N , are two independent, focused firms, and their shares are separately traded. We derive only the results for M , the results for N being the same except for the differences in notations.

Suppose that trader $\mathcal{I}$ acquires the information about ε for unit M . Generally, his strategy depends on the realization of ε . But so that he does not fully reveal the true realization, there has to be a common element in his state-dependent strategies. In general there are two (and only two) ways constructing a strategy that is not fully revealing:

- (1) If the informed trader's order in one state is b plus that in the other state, the total order flows in the state without a liquidity shock is the same as that in the other state with a liquidity shock (see the second and third rows in column 4 of Table 1 below).
- (2) The informed trader's order in each of the two states is the same.

The second way does not utilize available information and hence is not profitable. Therefore, we need only to look at the first kind of strategies. They can be written as the following generic form:¹⁶

S1-Strategies: Sell αb shares in the good state ($\varepsilon = e$) and sell $(1+\alpha)b$ shares in the bad state ($\varepsilon = -e$), where α is between -1 and 1 .

As one can see from Table 1 below, under a $S1$ strategy, the order flow when $\varepsilon = e$ and $\beta = b$ is $(1+\alpha)b$, which is the same as that when $\varepsilon = -e$ and $\beta = 0$. For example, when α is -1 , the informed

¹⁶We thank the referee for pointing out this general way of formulating the strategies; and, as it turns out, this class of strategies are all equilibrium strategies.

trader is buying b shares in the good state and trade no shares in the bad state. When $\alpha = 0$, he trades no shares in the good state and sells b shares in the bad state. Note that if the informed trader buys b shares in the good state and sells b shares in the bad state, the order flow will fully reveal his trades.

Table 1. Order flows, market prices, and investment decisions for business unit M

(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
ε	$\mathcal{I}$'s order	β	Order Flow	Market Price	Liquidation Decision	$\mathcal{I}$'s profit	Final Payoff
e	αb	0	αb	$m + e$	No	0	$m + e$
		b	$(1 + \alpha)b$	m	No	$-\alpha be$	$m + e$
$-e$	$(1 + \alpha)b$	0	$(1 + \alpha)b$	m	No	$(1 + \alpha)be$	$m - e$
		b	$(2 + \alpha)b$	$m - e$ or 0	No if $m \geq e$ Yes if $m < e$	0	$\max(m - e, 0)$

Column (1) and (3) of Table 1 are the realizations of ε and β . There are four possible realizations of (ε, β) : $(e, 0)$, (e, b) , $(-e, 0)$, and $(-e, b)$. The total order flow is the sum of the orders submitted by trader $\mathcal{I}$ and the liquidity sell order β . The market price is determined as in Kyle (1985). That is, the market makers observe the aggregate order flow and set the price equal to the expected payoff, which is conditioned on the information conveyed by the total order flow. The firm then uses the information conveyed by the market price to make its liquidation decision. When the price is m , the firm knows that the realization is either (e, b) or $(-e, 0)$ with equal probability. Since expected payoff is $m > 0$, the firm will not liquidate. However, if the market price is $m - e$ or 0, then the firm knows for sure that $\varepsilon = -e$. Its investment decision is then: liquidates if $m < e$ and does not liquidate if $m \geq e$.

Each of the four possible realizations of (ε, β) will happen with probability 0.25. Expected profit for the informed trader is $\frac{be}{4}$, the weighted average of the values in column (7). The expected payoff of the firm is $\frac{3m + \max(m, e)}{4}$, the weighted average of the values in column (8). The firm M 's value is the price investors are willing to pay at date 0. As in other models of this type, this value is the

difference between the expected payoff of the firm and the anticipated loss (uninformed) investors may have to incur when selling the shares at date 1 to the informed trader. This adverse selection discount represents a loss of value to the firm. Proposition 1 summarizes the discussion above.

Proposition 1 *Any S1 trading strategy is an equilibrium; and for any such strategy, the profit of the informed trader is $be/4$. The date 0 expected value of the firm M is $\frac{3m + \max(m, e)}{4} - \frac{be}{4}$.*

Proposition 1 shows that all S1-strategies are equilibrium strategies. The trading profit for the informed trader, however, remains the same under all the equilibria.

We now examine the investment efficiency under these equilibria. In Table 1, order flows partially reveal the realization of ε (column (4)). The market makers and the firm M are able to distinguish between the two states half of the time. Note that without learning from the market price, the expected payoff from investment is m . Now it is $\frac{3m + \max(m, e)}{4}$, which is greater than m if $m < e$. That is, there is an improvement in investment efficiency when the information about ε is valuable ($m < e$). On the other hand, if $m > e$, there is no improvement because the information adds no value to the business unit and is not needed.¹⁷

To focus on the most interesting cases, we make the following assumption:

Assumption 1 $\frac{bd}{4} - c \geq 0$.

Without loss of generality we also assume $e \geq d$. Therefore, Assumption 1 implies $\frac{be}{4} - c \geq 0$. Under Assumption 1, trader $\mathcal{I}$ has an incentive to gather and trade on information about both ε and δ . The share prices of firm M and firm N are therefore partially informative. Improvements in investment efficiency are realized in both firms.

¹⁷By replacing m with n and e with d , we obtain the results for N . Trader $\mathcal{I}$'s net trading profit is $\frac{bd}{4} - c$. The date 0 equilibrium value of N is $\frac{3n + \max(n, d)}{4} - \frac{bd}{4}$.

V_0^M and V_0^N denote the values of M and N as separately listed firms. Then the combined date 0 value of the two firms is

$$V_0^M + V_0^N = \frac{3m + \max(m, e)}{4} + \frac{3n + \max(n, d)}{4} - \frac{b(e + d)}{4} \quad (1)$$

4 Liquidity, Investment Efficiency and Value under Integration

In this section we study the case where the two business units are integrated and only the shares of the integrated firm are traded. We answer the question: if the two units merge, what type of information will the informed trader $\mathcal{I}$ gather? how does he trade?

In this case, trader $\mathcal{I}$ has three choices: (i) collects only the information about ε at cost of c ; or, (ii) collects only the information about δ at cost of c ; or, (iii) collects information of both ε and δ at cost of $2c$. Since by assumption $e \geq d$, information about ε is more valuable than δ , so (ii) is always dominated by (i). We will examine the informed trader's profit under each of the two remaining cases (case (i) and (iii)), and then derive the conditions that determine his choice of information acquisition.

4.1 Acquiring One Type of Information under Integration

We first consider case (i) where the informed trader collects only the information about ε . In this case, his trading strategies can only be contingent upon the realization of ε , and the class of the strategies that will not fully reveal his trades is the same as before: the $S1$ strategies.¹⁸ The following result shows that given that his choice is (i), his trading profit under integration is the same as that when the business unit M is separately traded.

¹⁸Strictly speaking, the $S1$ strategies under integration is not exactly the same as those defined in the previous section because there are more states due to the realization of δ . The similarity lies in the fact that they only depend on the realization of ε . For simplicity, we will use the same notation.

Proposition 2 *Suppose only information about ε is collected under integration. Then any S1 strategy is an equilibrium strategy. The equilibrium trading profit for trader $\mathcal{I}$ is $be/4$. The value of the integrated firm is $V_0^{M+N} = \frac{3m + \max(m, e)}{4} + n - \frac{be}{4}$.*

This proposition implies even when the two units are integrated, trading on ε only is always an equilibrium strategy. In this case, the stock price partially reveals information about ε , while the information about δ is fully suppressed. Consequently, the improvement of investment decisions can only be achieved in unit M , whose investment payoff is the same as in the spin-off case, $\frac{3m + \max(m, e)}{4}$. Since information about N is not revealed, the expected payoff of unit N is its unconditional NPV, n . The value of the integrated firm at time 0 is the sum of the payoffs of these two units minus the uninformed investors' expected loss to the informed trader $\mathcal{I}$.

4.2 Acquiring Two Types of Information under Integration

Next we consider case (iii) where both types of information are collected. Notice that the strategy space has been augmented because trading strategies can be made contingent on four possible states corresponding to the four possible realizations of the (ε, δ) pair. Again, in order to prevent trader $\mathcal{I}$'s trades from fully revealing, there has to be some kind of links between his trades in the four different states (this puts restrictions on trading strategies). Similar to our previous discussion, there are generally two ways to link (pool) any two of the four states: (1) the informed trader's order in one state is b plus that in the other state so that when the liquidity shock is absent, so the total order flow looks the same as that in the other state when the liquidity shock is present; (2) the informed trader's order in two states is the same.

We have listed all these possible strategies in Table 3 of the Appendix with their trading profits calculated. Note that even when a strategy yields a high profit, it may not be an equilibrium strategy

because trader $\mathcal{I}$ may be induced to deviate from it. The rational trader $\mathcal{I}$ will play these equilibrium strategies that give him the highest profit. In what follows, we will first find the strategies that give the highest profit to the informed trader. Then we will consider under what conditions they are equilibrium strategies.

By examining Table 3 in the Appendix, the strategies that yield the highest profit is of the following form (listed as strategy 4.2 in Table 3):

I1-Strategies: Sell $\alpha_1 b$ shares in state (e, d) , sell $(1 + \alpha_1)b$ shares in states $(e, -d)$ and $(-e, d)$, and sell $(2 + \alpha_1)b$ shares in state $(-e, -d)$.

The corresponding order flows under I1 strategies are illustrated in Table 4 of the Appendix. For example, when $\alpha_1 = -1$, this strategy calls for buy b shares in (e, d) , trade no shares in states $(e, -d)$ and $(-e, d)$, and sell b shares in state $(-e, -d)$. In this class of strategies, both state (e, d) and state $(-e, -d)$ are pooled with states $(e, -d)$ and $(-e, d)$ in the first way discussed above. And states $(e, -d)$ and $(-e, d)$ are pooled in the second way.

By comparing the potential informed trading profits under other candidate strategies illustrated in Table 3 of the Appendix to these produced by the I1 strategies, we have the following result:

Lemma 1 *Among all the possible strategies contingent on the realizations of both ε and δ , the highest possible trading profit for the informed trader is $\frac{b(e+d)}{6}$ generated by I1 strategies.*

Note the $\frac{b(e+d)}{6}$ is the gross profit for the informed trader. Since he pays to learn about both ε and δ , his net profit is $\frac{b(e+d)}{6} - 2c$.

To understand the intuition of why this type of strategies will yield the highest profit for trader $\mathcal{I}$ when both types of information are collected, let us first note that by Proposition 2, the informed

trading profit is $be/4$ if only the type ε information is collected. And in this ($S1$ strategy) equilibrium, trader $\mathcal{I}$ can only make his trade contingent on the realization of ε . Therefore, he buys the same amount in states (e, d) and $(e, -d)$ and sells the same amount in states $(-e, d)$ and $(-e, -d)$. Now when the information about δ becomes available, the strategies that can intuitively do better are those that will modify the buy and sell orders depending on whether the two types of information are offsetting or not. That is, these strategies should reduce the buy order in state $(e, -d)$ and reduce the sell order in state $(-e, d)$ if d is relatively important comparing to e (e is not too large relative to d) so that the two types of information are offsetting in states $(e, -d)$ and $(-e, d)$.

Proposition 3 gives the conditions under which an $I1$ strategy is an equilibrium strategy.

Proposition 3 *Suppose that information about both ε and δ is collected under integration. There exists a parameter region Ω in which an $I1$ strategy is an equilibrium. In this equilibrium, the informed trader's profit is $(e + d)b/6$. The time 0 value of the integrated firm is $V_0^{M+N} = \frac{7m + \max(m, e)}{8} + \frac{7n + \max(n, d)}{8} - \frac{b(e + d)}{6}$.*

Note that for some parameter values, no $I1$ strategy can be an equilibrium. The reason is that in those regions, it pays for the informed trader to deviate to other strategies. The definition of Ω is given at the end of the proof of Proposition 3 in Appendix. Essentially, Ω consists of regions (for different values of m and n) in which e is required not to be too large relative to d . If e is too large relative to d , the two types of information do not offset each other out in the two middle states $(e, -d)$ and $(-e, d)$. Rather than placing the same order in these two middle states as an $I1$ strategy does, it pays trader $\mathcal{I}$ to deviate and place the same order in state $(e, -d)$ as in state (e, d) .

Since $I1$ strategies yield the highest profit for trader $\mathcal{I}$, he will play it if it is an equilibrium. In the regions where an $I1$ strategy cannot be an equilibrium, we will consider the strategies from Table 3 which yield the next highest profit for trader $\mathcal{I}$ to see if they can be an equilibrium, and then the

strategies with the third highest profit. This process will continue until all the possibilities are covered. As we will see below, we do not need to do this because most of these strategies yield a lower profit for the informed trader than that when only one type of information is collected (strategy $S1$).

4.3 The Choice of Information Acquisition by $\mathcal{I}$ under Integration

The informed trader $\mathcal{I}$ is willing to play an equilibrium strategy that yields him the highest profit. To determine whether he will acquire information about ε only or about both ε and δ , we need to compare his trading profits under choice (i) and choice (iii).

Proposition 4 *The informed trader's decision to collect information under integration.*

- *There is a critical cost level c^* such that, if the actual information acquisition cost c is less than c^* and that parameter values fall in Ω , then trader $\mathcal{I}$ will collect information about both ε and δ and play the equilibrium strategy $I1$;*
- *Otherwise, trader $\mathcal{I}$ only collects information about ε and play the equilibrium strategy $S1$.*

The conditions in Proposition 4 are relatively easy to understand. When the information collection cost c is small, it pays to learn both ε and δ . Another factor is the relative importance of δ and ε . If d is very small relative to e , the informed trader is better off by just learning the more important information ε about unit M . Ω is the parameter region where an $I1$ strategy is an equilibrium one.

The second part of Proposition 4 also implies that even when $c < c^*$, as long as $I1$ fails to be an equilibrium (parameter values are not in Ω), then only information on ε will be collected. That is because other than strategy $I1$, only one strategy (strategy 4.3) in Table 3 of the Appendix *could possibly* yield an informed trading *gross* profit higher than that under $S1$ strategy. However, it can be shown that strategy $S1$ dominates strategy 4.3 in terms of net profit (see Lemma 2 in Appendix). Therefore,

in the regions where an $I1$ strategy cannot be an equilibrium, no other strategy based on both types of information will yield a higher payoff for the informed trader than that under $S1$. Consequently, the informed trader will only need to collect information about ε and play strategy $S1$.

4.4 Stock Price Efficiency and Investment Efficiency under a Merger

When the two business units are integrated (merged), according to Proposition 4, it is no longer optimal for trader $\mathcal{I}$ to *always* acquire and trade on both types of the information. Consequently, the informed trader's profit as well as the informational efficiency of stock prices are lower under a merger than under a spin-off. The lowered informational efficiency may result in allocative inefficiency in the business units' investment decisions.

From trader $\mathcal{I}$'s perspective, the integration makes him worse off. When the two business units are merged and when he acquires the knowledge of both ε and δ , his net profit becomes $\frac{b(e+d)}{6} - 2c$, which is less profitable compared to his informed trading profit under a spin-off: $\frac{b(e+d)}{4} - 2c$. This is because although in both cases he is paying $2c$, there is only one stock price for him to trade on and profit from. When he acquires only the knowledge about ε , his net gain is $\frac{be}{4} - c$ under the integration, which is again less profitable than what he can obtain under a spin-off.

The lower informed trading profit naturally leads to lower stock price efficiency. This is due to two reasons. First, the informed trader is more likely to trade on only one type of information. In this case, the information about the investment opportunity of division N is lost. Secondly, even if the informed trader acquires the knowledge on both ε and δ and trades upon them, with a single stock price it is more difficult to disentangle one type of information from the other. That is, the stock price will not be able to distinguish completely between the state in which ε is positive and δ is negative and the state in which ε is negative and δ is positive. This is especially the case when the two types of information

are almost equivalently important (the magnitude of δ , d , is relatively close to that of ε , e). Of course, whether this informational inefficiency will result in allocative inefficiency depends on the means and variances of the investment opportunities. When $n < d$, that is, when the information about unit N is socially valuable, not knowing it can lead to an investment with a NPV of $n - d < 0$. We will see this point more clearly in the next section.

The result that mergers lead to less informative stock prices and potentially less efficient investment decisions is consistent with much of the empirical evidence concerning internal capital markets in conglomerate firms. In our model, however, this potential loss in investment efficiency is offset by the lower liquidity discount in the stock returns.

5 Conglomerate Mergers versus Spin-offs

In the previous two sections, we derived the values of the two business units under spin-off (see (1)) and under integration (see Propositions 2 and 3). In this section, we first derive the conditions that determine when a merger or a spin-off is optimal and then identify what type of firms should be the candidates for a merger or a spin-off.

5.1 The Conditions of a Conglomerate Merger

Formally, a merger is optimal for the two business units if and only if

$$V_0^M + V_0^N \leq V_0^{M+N} \tag{2}$$

Propositions 5 presents detailed conditions under which a merger or a spin-off is optimal.

Proposition 5 *The conditions determining merger over spin-off are the follows:*

(1) Suppose $c \geq c^*$. The two business units will prefer merger over spin-off if and only if $n > (1 - b)d$.

In particular, merger is always optimal if $n > d$.

(2) Suppose $c < c^*$. If an *I1* strategy is an equilibrium, then the two business units prefer merger over spin-off if and only if $\frac{b(e + d)}{3} > \frac{\max(m, e) - m + \max(n, d) - n}{2}$; otherwise the condition for merger is the same as that in (1).

When $c \geq c^*$, only one type of information, ε , is collected under a merger. The first part of Proposition 5 is derived by comparing equation (1) with the value of the integrated firm in Proposition 2. To understand the condition $n > (1 - b)d$, note that when $n > d$, information about δ does not help to improve investment decisions because even in the worst state $\delta = -d$, the investment has an NPV of $n - d > 0$ and should be made. Therefore, no socially valuable information is lost in a merger. When $n < d$, this is not the case. Inefficient investment can happen after a merger. However, if the fraction of investors who will experience liquidity shocks b is relatively large, a merger can reduce their losses to the informed trader when they have to sell. Therefore, a merger can still be optimal.

When $c < c^*$, the informed trader $\mathcal{I}$ will acquire information about both ε and δ if an *I1* strategy is an equilibrium (parameter values fall in Ω). The second part of Proposition 5 is then obtained by comparing equation (1) with the firm's value derived in Proposition 3. When d is relatively large and/or the cost of learning c is small ($c < c^*$), it is possible to induce the informed trader to collect information about both ε and δ after a merger and play the equilibrium strategy *I1*. The value of the firm in this case is $V_0^{M+N} = \frac{7m + \max(m, e)}{8} + \frac{7n + \max(n, d)}{8} - \frac{b(e + d)}{6}$. Comparing it with the value under spin-off $V_0^M + V_0^N = \frac{3m + \max(m, e)}{4} + \frac{3n + \max(n, d)}{4} - \frac{b(e + d)}{4}$, it is clear that a merger between the two business units creates greater value than a spin-off if V_0^{M+N} is greater than $V_0^M + V_0^N$. That is,

$$\frac{b(e + d)}{3} > \frac{\max(m, e) - m + \max(n, d) - n}{2}$$

Notice that this condition depends solely on the “mean return” (m and n) and the “variance” (ε and δ) of the investment opportunities of the two business units.

5.2 Identifying Merger and Spin-off Candidates

We are now going to address the following questions: What type of firms are most likely to be involved in conglomerate mergers? What type of firms are most likely to be spun off? To answer these questions, we need to classify firms along the means and variances of their investment opportunities. To do this, we impose the following ordering on the parameters:

Assumption 2. $e > m > d > n \geq 0$.

The ordering in Assumption 2 enables us to obtain a simple classification of business units. We can characterize the real-world business units into four types by pairing up two possible means (m and n) and two possible volatilities (ε and δ) of their investment opportunities:

- Type 1 (the high return and high risk type): (m, ε) ;
- Type 2 (the high return and low risk type): (m, δ) ;
- Type 3 (the low return and high risk type): (n, ε) ;
- Type 4 (the low return and low risk type): (n, δ) .

We consider a merger (or a spin-off) between any two of these four types of firms. To simplify the analysis, we also make the assumption which allows us to focus on the case where only information on ε is collected after a merger.¹⁹

¹⁹The analysis when this assumption is violated can be similarly carried out. The only difference is to use the firm’s value under integration in Proposition 3 rather than that in Proposition 2 in the computation below. It will not be presented here due to space limitation.

Assumption 3. $c \geq c^*$.

Let i or j be the index of a firm's type. Define the gain/loss of merger over spin-off between a type i firm and a type j firm as the difference between the merged value of i and j minus the sum of the stand-alone values of i and j :

$$\Delta_{ij} = V_0^{i+j} - (V_0^i + V_0^j) \quad (3)$$

These merger gains (or losses if negative) between any two types of units, Δ_{ij} , are calculated based on (3) and presented in Table 2 below.

Table 2. The gains/losses of a merger over a spin-off between any two types of business units

	(m, ε)	(m, δ)	(n, ε)	(n, δ)
(m, ε)	$\frac{m - (1 - b) e}{4}$	$\frac{bd}{4}$	$\frac{n - (1 - b) e}{4}$	$\frac{n - (1 - b) d}{4}$
(m, δ)	$\frac{bd}{4}$	$\frac{bd}{4}$	$\frac{bd}{4}$	$\frac{bd}{4}$
(n, ε)	$\frac{n - (1 - b) e}{4}$	$\frac{bd}{4}$	$\frac{n - (1 - b) e}{4}$	$\frac{n - (1 - b) d}{4}$
(n, δ)	$\frac{n - (1 - b) d}{4}$	$\frac{bd}{4}$	$\frac{n - (1 - b) d}{4}$	$\frac{n - (1 - b) d}{4}$

The cell with the largest value corresponds to the two types of firms between which a conglomerate merger is most likely to occur. The cell with the smallest value corresponds to the two types of units that are most likely to be split up (or remain independent). Table 2 provides us with a guide.

If the merger gains in a column of Table 2 are uniformly greater than or equal to those in another column, then the type of firms represented by the former column is more likely to be involved in conglomerate mergers than the type of firms represented by the latter column, other things being equal. This result is summarized in Proposition 6.

Proposition 6 *Other things being equal, the low-return-high-risk type firm is most likely to be focused (or spun off). The high-return-low-risk type firm is most likely to be involved in conglomerate mergers if $m < (1 - b)e + bd$.*

Intuitively, the low-return-high-risk type firm stands to gain the most from a more efficient investment decision. Staying focused helps to improve its decision through more information revelation. The converse is true for the high-return-low-risk type firm. Specifically, only for the former type of firm, information acquisition generates any value for the firm, since the information observed by the informed trader guides the capital allocation process. In the latter case, even the stock price becomes more informative, it does not affect the capital allocation decisions of the firm. Because $m < e$, the condition in the second part of Proposition 6 holds when the liquidity shock b is not too large.

During a business' life cycle, the mean and variance of its investment opportunities change. According to Proposition 6, its likelihood of being involved in conglomerate mergers or spin-offs also changes. Many businesses start out as a type 1 firm after their IPOs due to high uncertainty of their growth potential (internet firms for example). If this uncertainty does not make them a failure, they often become the leader in their industries (such as Intel and Microsoft) with high profitability. This changes them to a type 2 firm. As more competitors enter into their industries and their profitability drops. They turn into a type 3 or 4 firm (such as IBM or Kodak). Our theory would predict that the businesses following this life cycle are likely to start out as focused firms, then to make mergers and acquisitions, and finally to break up.

6 Conclusion

Discussions on the economic efficiency of conglomerate mergers and spin-offs have invariably focused on the issues related to information. In this paper, we establish a link between the first two moments of firms' investment opportunities and the informational benefits and costs of mergers. We focus on the information that managers either do not have or would like to suppress due to agency problems when they make investment decisions. By explicitly building firms' investment decisions into a market microstructure model, we have developed a theory of conglomerate mergers in which both the benefits and the costs of diversification are related to the characteristics of firms' investment opportunities. The benefits come from a reduced liquidity discount in the stock price of the merged firm when its shareholders anticipate less informed trading in the secondary market. The costs are the result of less efficient investment by the firm's divisions due to a less informative stock price. The benefits and the costs vary with the means and the variances of the NPVs of the investment opportunities of the firm's divisions. As the investment opportunities change over time, mergers or spin-offs will occur in response to the changing benefits and costs. For two business units with different types of investment opportunities, we can explicitly compute the gain or loss of a merger or a spin-off as a function of the means and variances of their investment opportunities. Our results show that it is important to distinguish between the risk and the return of the firm's investment opportunities when talking about mergers. Our results also have implications for evaluating merger or spin-off candidates and for empirical work on corporate restructuring.

A few assumptions of the model can be relaxed without qualitatively changing our results. One such assumption is that a single informed trader learns both types of information. In the previous version of this paper, we also analyzed the case where there are two informed traders and each one can obtain information on one division. The results are the same except the right hand side of the condition in

Assumption 3 is slightly different.²⁰

A number of issues remain unanswered. For example, does a financial intermediary (such as a mutual fund) have incentives to buy shares of two separately traded business units and issue a composite security? If it does, what would be the impacts on liquidity, information revelation and investment efficiency? We know that if the intermediary buys all the outstanding shares of the two firms, the effect is exactly like a merger. However, if it only buys a fraction of shares, this may produce an intermediate outcome between a spin-off and a merger. An analysis along this line should be interesting.²¹ Another related issue is capital structure. Can trading of both debt and equity achieve the same efficiency as trading of two equities? These are interesting issues for future research. Our paper is only a first step in that direction.

7 Appendix.

Proof of Proposition 1. See the text. ■

Proof of Proposition 2. One can construct a table similar to Table 1 with the realizations of δ added as another column (the number of rows will double due to the two realizations of δ). Since only the information about ε is collected, the informed trader's strategy is still contingent on the realizations of ε (not those of δ) and the aggregate order flows are the same as in Table 1. The only difference is that the expected value of the second division, n , is added to all the market prices and final payoffs. The rest of proof follows that of Proposition 1. ■

²⁰The condition becomes $c \geq c^{**}$, where $c^{**} \equiv \frac{b(3d-e)}{8}$. The right hand side is higher reflecting the fact the the informed trader who has the access to the type δ information is more likely to do that when the two informed traders do not coordinate their strategies. We will provide these results to the interested readers upon request.

²¹ Financial intermediation in the forms of diversified stock mutual funds and stock index trading may replicate some of the benefits of conglomerate mergers. The rapid development of these instruments in the U.S. may be partially responsible for the spin-off wave in the recent decades.

Proof of Lemma 1. When both types of information are observed, the informed trader's strategy will be contingent on the four possible realizations of (ε, δ) . In order to make his trading profitable, his equilibrium strategies should not fully reveal all of the four states. There are only two ways in which the informed trader can prevent his trade from being fully revealed: (i) his order in one state is b plus that in the other state so that the total order flow in the first state without the liquidity shock looks the same as that in the second state with the liquidity shock; (ii), his order in two states is the same.

In Table 3, all possible strategies that are not fully revealing are listed lexicographically (see Remark 1 after this proof). Based on Table 3, we then compute the informed trader's *potential* trading profit for each of these strategies. The calculations are similar to these in Proposition 1. Note that for some strategies, the informed trading profit varies in different parameter region. That is, the trading profit depends on the relative magnitude between m (n) and e (d).

By comparing these trading profits, the highest profit is the one generated by strategy 4.2 (strategy I1 in the text). ■

Remark 1 *How the strategies are lexicographically listed:*

We start with a strategy in which the informed's sell order in each state is the same αb for an arbitrary α (strategy 1.1). In the next two rows, we change the informed's order in the last state $(-e, -d)$ to $(1 + \alpha)b$ to get strategy 1.2 and then to $\alpha_1 b$ for an arbitrary α_1 to get strategy 1.3.

The next class of strategies involve changing the informed order in the state 3 $(-e, d)$ to $(1 + \alpha)b$ and then vary the order in state 4 $(-e, -d)$ as before. Now there are four general ways to vary the order in state 4 because $(2 + \alpha)b$ should be treated as separate case because it is the order in state 3 plus αb .

The third class of strategies has the sell order changed to α_1 in state 3, where α_1 is neither α nor $(1 + \alpha)b$. The fourth and fifth classes of strategies are derived similarly.

It should be pointed out now that since the four states in the first row of Table 3 are ordered from the best (highest payoff on the projects) to the worst, the informed trader's sell order should not go down as we move from the left to the right in order to make higher profit. That is, a strategy such as 2.1 is always dominated by a strategy such as 2.2 in which the sell order in state 4 is at least as high as that in state 3. In the listing below the third class, we will not list those dominated strategies.

Table 3. Informed trading strategies and profits ($\alpha_1 \neq \alpha, 1 + \alpha, 2 + \alpha$)

	(e, d)	$(e, -d)$	$(-e, d)$	$(-e, -d)$	Informed Trading Profits
1.1	αb	αb	αb	αb	0
1.2	αb	αb	αb	$(1 + \alpha) b$	$\frac{(e+d)b}{8}$
1.3	αb	αb	αb	$\alpha_1 b$	0
2.1	αb	αb	$(1 + \alpha) b$	αb	$\frac{(e-d)b}{8}$
2.2	αb	αb	$(1 + \alpha) b$	$(1 + \alpha) b$	$\frac{eb}{4}$
2.3	αb	αb	$(1 + \alpha) b$	$\alpha_1 b$	$\frac{(2e-d)b}{12}$
2.4	αb	αb	$(1 + \alpha) b$	$(2 + \alpha) b$	$\frac{(4e+d)b}{24}$
3.1	αb	αb	$\alpha_1 b$	$\alpha_1 b$	0
3.2	αb	αb	$\alpha_1 b$	αb	0
3.3	αb	αb	$\alpha_1 b$	$(1 + \alpha) b$	$\frac{(2e+d)b}{12}$
4.1	αb	$(1 + \alpha) b$	$(1 + \alpha) b$	$(1 + \alpha) b$	$\frac{(e+d)b}{8}$
4.2	αb	$(1 + \alpha) b$	$(1 + \alpha) b$	$(2 + \alpha) b$	$\frac{(e+d)b}{6}, \frac{(e+d)b}{12}, \frac{(2e+d)b}{12}, \frac{(e+2d)b}{12}$
4.3	αb	$(1 + \alpha) b$	$(2 + \alpha) b$	$(2 + \alpha) b$	$\frac{(4e+3d)b}{24}, \frac{(4e+d)b}{24}, \frac{db}{24}$
4.4	αb	$(1 + \alpha) b$	$(2 + \alpha) b$	$(3 + \alpha) b$	$\frac{(e+d)b}{8}$
5.1	αb	$\alpha_1 b$	$\alpha_1 b$	$\alpha_1 b$	0
5.2	αb	$\alpha_1 b$	$\alpha_1 b$	$(1 + \alpha_1) b$	0
5.3	αb	$\alpha_1 b$	$(1 + \alpha_1) b$	$(1 + \alpha_1) b$	0
5.4	αb	$\alpha_1 b$	$(1 + \alpha_1) b$	$(2 + \alpha_1) b$	0

Proof of Proposition 3.

We derive the conditions under which an *I1* strategy (that is, strategy 4.2 in Table 3) becomes an

equilibrium. Since we have to check many possible deviations under different parameter values, the proof is quite long.

Step 1: Deriving prices and investment decisions when $I1$ is played. An $I1$ strategy is illustrated in the second column of Table 4, with α varies from -1 to 1 . The aggregate order flow observed by the market maker is the sum of informed trader's order and the liquidity traders' orders (See column 4 in Table 4).

Table 4.

(1)	(2)	(3)	(4)
Realization of (ε, δ)	Informed Order	Liquidity Order β	Aggregate Order Flow
(e, d)	αb	0	αb
		b	$(1 + \alpha) b$
$(e, -d)$	$(1 + \alpha) b$	0	$(1 + \alpha) b$
		b	$(2 + \alpha) b$
$(-e, d)$	$(1 + \alpha) b$	0	$(1 + \alpha) b$
		b	$(2 + \alpha) b$
$(-e, -d)$	$(2 + \alpha) b$	0	$(2 + \alpha) b$
		b	$(3 + \alpha) b$

The posterior probability that state (e, d) has occurred upon observing the aggregate order flow, αb , is $\Pr(e, d | \alpha b) = 1$. The order flow fully reveals the true state, therefore, the quoted price is $m + e + n + d$. The firm will invest in both M and N upon seeing this price.

On the other hand, the order flow $(1 + \alpha) b$ can be generated by three states, so the posterior is

$\Pr(e, d | (1 + \alpha)b) = \Pr(e, -d | (1 + \alpha)b) = \Pr(-e, d | (1 + \alpha)b) = 1/3$. The order flow is not revealing, so the quoted price is $m + n + \frac{e+d}{3}$. The firm will also invest in both M and N because based on the posterior the expected payoff with investment is higher than that without investment.

When order flow $(2 + \alpha)b$ is observed, the posterior probability is $\Pr(e, -d | (2 + \alpha)b) = \Pr(-e, d | (2 + \alpha)b) = \Pr(-e, -d | (2 + \alpha)b) = 1/3$. In this case, the expected value of the firm, if investments are made in both divisions, is $m + n - \frac{e+d}{3}$. Given the posterior, the investment in M should be made if and only if $m \geq \frac{e}{3}$; and investment in N should be made if and only if $n \geq \frac{d}{3}$. The price set by the market maker will rationally anticipate the firm's optimal investment decision after it sees the price. Therefore, the price will be $m + n - \frac{e+d}{3}$ if $m \geq \frac{e}{3}$ and $n \geq \frac{d}{3}$; $m - \frac{e}{3}$ if $m \geq \frac{e}{3}$ and $n < \frac{d}{3}$; $n - \frac{d}{3}$ if $n \geq \frac{d}{3}$ and $m < \frac{e}{3}$; and 0 if $m < \frac{e}{3}$ and $n < \frac{d}{3}$.

When the order flow $(3 + \alpha)b$ is observed, $\Pr(-e, -d | (3 + \alpha)b) = 1$. The expected value of the firm, if both divisions invest, is $m - e + n - d$. Again, investment in M is made if and only if $m \geq e$. Investment in N is made if and only if $n \geq d$.

In Table 5 we list the stock price and investment decision for different parameter values of the model. The table only shows the possible price and investment decision for order flows $(2 + \alpha)b$ and $(3 + \alpha)b$. We omitted the cases when the total order flow is αb or $(1 + \alpha)b$ because in those cases investments in M and N are always made and the prices are $m + e + n + d$ and $m + n + \frac{e+d}{3}$, respectively.

Table 5.

	Parameter Values	Price when order	Investment	Price when order	Investment
		flow is $(2 + \alpha) b$	at $(2 + \alpha) b$	flow is $(3 + \alpha) b$	at $(3 + \alpha) b$
(1)	$m \leq \frac{e}{3}, n \leq \frac{d}{3}$	0	No	0	No
(2)	$\frac{e}{3} \leq m \leq e, n \leq \frac{d}{3}$	$m - \frac{e}{3}$	M	0	No
(3)	$m \leq \frac{e}{3}, \frac{d}{3} \leq n \leq d$	$n - \frac{d}{3}$	N	0	No
(4)	$\frac{e}{3} \leq m \leq e, \frac{d}{3} \leq n \leq d$	$m + n - \frac{e+d}{3}$	M, N	0	No
(5)	$m \leq \frac{e}{3}, n \geq d$	$n - \frac{d}{3}$	N	$n - d$	N
(6)	$\frac{e}{3} \leq m \leq e, n \geq d$	$m + n - \frac{e+d}{3}$	M, N	$n - d$	N
(7)	$m \geq e, n \geq d$	$m + n - \frac{e+d}{3}$	M, N	$m - e + n - d$	M, N
(8)	$m \geq e, n \leq \frac{d}{3}$	$m - \frac{e}{3}$	M	$m - e$	M
(9)	$m \geq e, \frac{d}{3} \leq n \leq d$	$m + n - \frac{e+d}{3}$	M, N	$m - e$	M

Step 2: Deriving the equilibrium profit in each state. The informed trader's equilibrium expected trading profit, π^e , can be expressed as

$$\pi^e = \sum_{\varepsilon, \delta} \pi^e(\varepsilon, \delta) \Pr(\varepsilon, \delta) = \sum_{\varepsilon, \delta} \left(\sum_{\beta} \pi^e(\varepsilon, \delta | \beta) \Pr(\beta) \right) \Pr(\varepsilon, \delta)$$

where $\pi^e(\varepsilon, \delta)$ is the equilibrium trading profit in state (ε, δ) . For a given state, $\pi^e(\varepsilon, \delta) = \frac{1}{2}\pi^e(\varepsilon, \delta | \beta = 0) + \frac{1}{2}\pi^e(\varepsilon, \delta | \beta = b)$. Table 6 lists the trading profit in each of the four states for different parameter values.

Table 6.

Case	$\pi^e(e, d)$	$\pi^e(e, -d)$	$\pi^e(-e, d)$	$\pi^e(-e, -d)$	π^e
(1) $m \leq \frac{e}{3}, n \leq \frac{d}{3}$	$-\frac{(e+d)\alpha b}{3}$	$\frac{(-e+2d)(1+\alpha)b}{3}$	$\frac{(2e-d)(1+\alpha)b}{3}$	0	$\frac{(e+d)b}{12}$
(2) $\frac{e}{3} \leq m \leq e, n \leq \frac{d}{3}$	$-\frac{(e+d)\alpha b}{3}$	$\frac{(-3e+2d)(1+\alpha)b}{3}$	$\frac{(3e-d)(1+\alpha)b}{3}$	$\frac{e(2+\alpha)b}{3}$	$\frac{(2e+d)b}{12}$
(3) $m \leq \frac{e}{3}, \frac{d}{3} \leq n \leq d$	$-\frac{(e+d)\alpha b}{3}$	$\frac{(-e+3d)(1+\alpha)b}{3}$	$\frac{(2e-3d)(1+\alpha)b}{3}$	$\frac{d(2+\alpha)b}{3}$	$\frac{(e+2d)b}{12}$
(4) $\frac{e}{3} \leq m \leq e, \frac{d}{3} \leq n \leq d$	$-\frac{(e+d)\alpha b}{3}$	$(-e+d)(1+\alpha)b$	$(e-d)(1+\alpha)b$	$\frac{(e+d)(2+\alpha)b}{3}$	$\frac{(e+d)b}{6}$
(5) $m \leq \frac{e}{3}, n \geq d$	$-\frac{(e+d)\alpha b}{3}$	$\frac{(-e+3d)(1+\alpha)b}{3}$	$\frac{(2e-3d)(1+\alpha)b}{3}$	$\frac{d(2+\alpha)b}{3}$	$\frac{(e+2d)b}{12}$
(6) $\frac{e}{3} \leq m \leq e, n \geq d$	$-\frac{(e+d)\alpha b}{3}$	$(-e+d)(1+\alpha)b$	$(e-d)(1+\alpha)b$	$\frac{(e+d)(2+\alpha)b}{3}$	$\frac{(e+d)b}{6}$
(7) $m \geq e, n \geq d$	$-\frac{(e+d)\alpha b}{3}$	$(-e+d)(1+\alpha)b$	$(e-d)(1+\alpha)b$	$\frac{(e+d)(2+\alpha)b}{3}$	$\frac{(e+d)b}{6}$
(8) $m \geq e, n \leq \frac{d}{3}$	$-\frac{(e+d)\alpha b}{3}$	$\frac{(-3e+2d)(1+\alpha)b}{3}$	$\frac{(3e-d)(1+\alpha)b}{3}$	$\frac{e(2+\alpha)b}{3}$	$\frac{(2e+d)b}{12}$
(9) $m \geq e, \frac{d}{3} \leq n \leq d$	$-\frac{(e+d)\alpha b}{3}$	$(-e+d)(1+\alpha)b$	$(e-d)(1+\alpha)b$	$\frac{(e+d)(2+\alpha)b}{3}$	$\frac{(e+d)b}{6}$

For example, to compute $\pi^e(-e, -d)$ for case 5, notice that the price for an observed total order flow $(2+\alpha)b$ (when the liquidity trader submit 0 order) is $n - \frac{d}{3}$, and for an aggregate order flow $(3+\alpha)b$ is $n - d$ (see Table 5, row (5)). So his expected profit in this state is

$$\pi^e(-e, -d) = \sum_{\beta} \pi^e(\varepsilon, \delta | \beta) \Pr(\beta) = \frac{(2+\alpha)b}{2} \left[n - \frac{d}{3} - (n-d) + n-d - (n-d) \right] = \frac{d(2+\alpha)b}{3}$$

The rest of expected equilibrium profit in the table are calculated similarly.

The informed trader's profit has to satisfy $\pi^e(\varepsilon, \delta) \geq 0$ for all ε and δ . Under I1, $\pi^e(e, -d) = -\pi^e(-e, d)$. This implies $\alpha = -1$, and $\pi^e(e, -d) = \pi^e(-e, d) = 0$. Accordingly, $\#\{I1 \text{ strategies}\} = 1$.

Step 3: Deriving the conditions under which deviations will not occur. The remaining of the proof derives the conditions that guarantee this strategy is an equilibrium strategy by ruling out

any profitable deviations.

Generally, a deviation is undetectable if the resulting order flow is αb , $(1 + \alpha) b$, $(2 + \alpha) b$, or $(3 + \alpha) b$. If a deviation result in an order flow other than these (that is, out of equilibrium path), the market maker will assume that the true state is the one in which the informed trader will benefit the most from the order. For example, if the market maker sees an order flow of $(4 + \alpha) b$, he will assume that the true state is $(-e, -d)$. This belief will prevent any deviation that results in an order flow that is out of equilibrium path. It is also straight forward to verify that the informed trader will not deviate from the current strategy in the state of (e, d) and $(-e, -d)$ because these are the two states in which he makes the highest profit: a deviation only reduces his profit.

In the state of $(e, -d)$ or $(-e, d)$, $\pi^e(e, -d) = \pi^e(-e, d) = 0$. The informed trader will deviate from it whenever a profitable trade is possible. In the case 2, 3, 5, and 8 of Table 6, the *I1* strategy cannot be an equilibrium strategy because the informed trader's deviation can generate a positive profit. For example, in case 2, if the informed trader learns the realization of the state is $(-e, d)$, he can submit an order of $(2 + \alpha) b$ instead of $(1 + \alpha) b$. The possible observed aggregate order flow, given that the liquidity investors will submit either 0 or b order, will be either $(2 + \alpha) b$ or $(3 + \alpha) b$. The market maker cannot detect the deviation and quotes the price of $m - \frac{e}{3}$ or 0 accordingly. The informed trader's expected trading profit due to the deviation will be (we use the superscript d to denote a deviation):

$$\pi^d(-e, d) = \frac{(2 + \alpha) b}{2} \left[m - \frac{e}{3} - (m - e) + 0 \right] = \frac{e(2 + \alpha) b}{3} > 0 = \pi^e(-e, d)$$

because $\alpha = -1$. Therefore, for the parameter region defined by case 2, *I1* cannot be an equilibrium strategy. Similarly, the informed trader can deviate to achieve a positive profit by submitting an order

flow of $(2 + \alpha)b$ instead of $(1 + \alpha)b$ when the state $(e, -d)$ occurs in case 3 and 5, and when the state $(-e, d)$ occurs in case 8.

We now discuss the remaining cases illustrated in Table 6.

In case 1, the parameter region is $m \leq \frac{e}{3}, n \leq \frac{d}{3}$. If the true state is $(e, -d)$, the informed trader can deviate by submitting an order of αb , $(2 + \alpha)b$ or $(3 + \alpha)b$. Since upon observing a total order flow of $(2 + \alpha)b$, $(3 + \alpha)b$ or $(4 + \alpha)b$, the market maker and the firm conjecture that the true states are among $(e, -d)$, $(-e, d)$, and $(-e, -d)$, no investment will be made and the prices given these order flows are zero (See Table 5, row (1)). Therefore, the informed trader will not deviate to $(2 + \alpha)b$ or $(3 + \alpha)b$. If he submits an order of αb , his profit due to deviation is

$$\pi^d(e, -d) = \frac{\alpha b}{2} \left[m + e + n + d - (m + e + n - d) + m + m + \frac{e + d}{3} - (m + e + n + d) \right] = \frac{(5d - e)\alpha b}{3}$$

Here, $\pi^d(e, -d) \leq 0$ if and only if $e \leq 5d$ because $\alpha = -1$.

On the other hand, if the realization of the state is $(-e, d)$, it can be shown that a deviation by αb gives the informed trader a negative profit of $\frac{(5e - d)\alpha b}{3}$ since $\alpha = -1$ and $e \geq d$. Deviation to $(2 + \alpha)b$ and $(3 + \alpha)b$ in this state will again generate zero profit since no investment will be made upon observing the two order flows. To summarize, strategy *I1* is an equilibrium strategy if $m \leq \frac{e}{3}, n \leq \frac{d}{3}, e \leq 5d$.

In case 4, the parameters satisfy $\frac{e}{3} \leq m \leq e, \frac{d}{3} \leq n \leq d$. When state $(e, -d)$ occurs, the deviation by αb can generate a $\pi^d(e, -d) = \frac{(5d - e)\alpha b}{3}$. For this deviation to be unprofitable, we need $\pi^d(e, -d) \leq 0$, or $e \leq 5d$. If state $(-e, d)$ occurs, informed trader can deviate by submitting an order of $(2 + \alpha)b$ instead of $(1 + \alpha)b$ (deviation by αb or $(3 + \alpha)b$ is not profitable in this state). This generates two possible order flows: $(2 + \alpha)b$ or $(3 + \alpha)b$. The expected trading profit from his deviation is $\pi^d(-e, d) = \frac{(2 + \alpha)b(e - 2d)}{3}$. Here, $e \leq 2d$ will guarantee $\pi^d(-e, d) \leq 0$ and that the deviation is unprofitable. In summary, strategy

$I1$ is an equilibrium strategy if $\frac{e}{3} \leq m \leq e, \frac{d}{3} \leq n \leq d$ and $e \leq 2d$.

In case 6, the parameters satisfy $\frac{e}{3} \leq m \leq e, n \geq d$. If the true state is $(e, -d)$, informed trader can deviate by submitting αb instead of $(1 + \alpha)b$ to achieve a profit of $\frac{(5d-e)\alpha b}{3}$. Again, $\pi^d(e, -d) \leq 0$ if and only if $e \leq 5d$ because $\alpha = -1$. The informed trader's deviation by submitting an order of $(2 + \alpha)b$ will lead to a negative profit of $\frac{(-2e+d)(2+\alpha)b}{3}$. Deviation by $(3 + \alpha)b$ results in a zero profit. Similarly, if the true state is $(-e, d)$, deviation by αb or $(3 + \alpha)b$ is not profitable. However, deviation by $(2 + \alpha)b$ generate $\pi^d(-e, d) = \frac{(e-5d)(2+\alpha)b}{3}$, which is non-positive if and only if $e \leq 5d$ when $\alpha = -1$. To summarize, strategy $I1$ is an equilibrium strategy if $\frac{e}{3} \leq m \leq e, n \geq d$ and $e \leq 5d$.

In case 7 where $m \geq e, n \geq d$, the possibly profitable deviation at $(e, -d)$ is still αb and at $(-e, d)$ is $(2 + \alpha)b$. Again, the condition $e \leq 5d$ is sufficient and necessary to guarantee these two deviations are unprofitable. That is, strategy $I1$ is an equilibrium strategy for the parameter region $m \geq e, n \geq d$ and $e \leq 5d$.

In case 9 where $m \geq e, \frac{d}{3} \leq n \leq d$, to rule out the only possibly profitable deviation at $(e, -d)$, αb , it requires $e \leq 5d$. If the true state is $(-e, d)$ instead, the possibly profitable deviation is $(2 + \alpha)b$ from $(1 + \alpha)b$. The corresponding aggregate order flow is either $(2 + \alpha)b$ (when $\beta = 0$) or $(3 + \alpha)b$ (when $\beta = b$). So the expected trading profit due to deviation in this state, based on the corresponding prices (see Table 5, row (9)), is $\pi^d(-e, d) = \frac{(2+\alpha)b(e-2d)}{3}$. To rule out a profitable deviation, we need $\pi^d(-e, d) \leq 0$ or $e \leq 2d$. In other words, the strategy is an equilibrium strategy for the parameter region $m \geq e, \frac{d}{3} \leq n \leq d$ and $e \leq 2d$.

To summarize the results in the previous paragraphs, an $I1$ strategy is an equilibrium strategy if the parameter values fall in one of the following regions: $\{m \geq \frac{e}{3}, \frac{d}{3} \leq n \leq d, e \leq 2d\}$, $\{m \leq \frac{e}{3}, n \leq \frac{d}{3}, e \leq 5d\}$, and $\{m \geq \frac{e}{3}, n \geq d, e \leq 5d\}$. Note that these conditions essentially require that e be not too large relative to d for different values of m and n . Let us define Ω to be the union of $\{m \geq \frac{e}{3}, \frac{d}{3} \leq n \leq d, e \leq 2d\}$,

$\{m \leq \frac{e}{3}, n \leq \frac{d}{3}, e \leq 5d\}$, and $\{m \geq \frac{e}{3}, n \geq d, e \leq 5d\}$. This completes the proof. ■

Lemma 2 *Strategy S1 dominates Strategy 4.3 of Table 3 when $m > \frac{e}{3}$ and $n < \frac{d}{3}$.*

Proof of Lemma 2. If the informed trader plays strategy 4.3, he can obtain a gross trading profit of $\frac{4e+3d}{24}b$ when $m > \frac{e}{3}$ and $n < \frac{d}{3}$.²² However, for this strategy to be an equilibrium strategy in this region, it requires $\pi^e(\varepsilon, \delta) \geq 0$ for all possible values of ε and δ . Otherwise the informed trader has the incentive to deviate. This requirement implies $-2 \leq \alpha \leq -1$ and $e \geq \frac{3}{2}d$.

To show strategy S1 dominates 4.3, it suffices to show that $\frac{be}{4} - c \geq \frac{4e+3d}{24}b - 2c$. But this is equivalent to show $\frac{2e-3d}{24}b \geq -c$. This condition is satisfied since $c \geq 0$ and we know from above the strategy requires $e \geq \frac{3}{2}d$. ■

Proof of Proposition 4. To derive the critical value of c^* , note that the net informed trading profit for trader $\mathcal{I}$ when only information about ε is collected is $\frac{be}{4} - c$ and the maximal profit for him when both types of information are collected is $\frac{b(e+d)}{6} - 2c$. The informed trader will collect only one type of information when $\frac{be}{4} - c \geq \frac{b(e+d)}{6} - 2c$, or $c \geq c^*$ where $c^* \equiv \frac{b(2d-e)}{12}$. If $c \geq c^*$, learning ε only and trading on it produces a higher net profit for trader $\mathcal{I}$ than learning about both ε and δ . If $c < c^*$, and as long as one of the existent conditions of Ω holds, learning about both ε and δ yields him a higher profit. So he will acquire both types of the information. If $c < c^*$ and none of the conditions defined in Ω holds, based on Lemma 2 and Table 3, learning both types of information is less profitable comparing to collecting the information about ε alone; so he will choose to acquire information about ε only. ■

Proof of Proposition 5. See the text. ■

Proof of Proposition 6. See the text. ■

²²Beyond this region, the possible informed trading profit is smaller than $\frac{4e+3d}{24}b$ and is strictly dominated by those under strategy I1 and S1. This can be verified by using Table 3. Detailed derivation and proof are available upon request.

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